

Implant, Conceal, Resurrect: The Fate of Inserted Skills Under Continued Training, from Toy Automata to a Language Model

Gunner Levi Howe
gunnerlevihowe@gmail.com

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Abstract

Can a skill be *inserted* into a model and survive continued training — and what protects it, at what plasticity cost? The factual-editing literature measures how key–value edits decay under fine-tuning, but without a ground-truth circuit it cannot distinguish a skill that is *erased* from one that is *bypassed*. We supply the missing instrument: a verified-exact S5 word-problem automaton constructed in a Householder linear RNN, implanted into a vocab-extended model and subjected to 60 epochs of continued training on mixtures of its own task and a foreign task, across four protection arms and three seeds (48 pre-registered cells). Three of our four frozen predictions died, and the structure that replaced them is sharper. First, protection is an *interaction*: removing the additive input path and giving the skill any exercise trickle (10% suffices) retains the implant perfectly (1.0 at every nonzero mixture), while the field-default architecture loses it even at 100% exercise (0.2467). Second — the resurrection — “dead” exercised implants are not dead: zeroing the additive projection at inference restores integrity to exactly 1.000 in 6/6 cells, including with every parameter trainable; training grew a parasitic bypass *around* an intact circuit, and the bypass’s norm scales with exercise. Third, the resulting fate taxonomy — **protected** / **concealed-but-recoverable** / **eroded** (true erosion only when unprotected *and* unexercised) — implies the edit-durability literature’s decay measurements may conflate two mechanistically different outcomes, only one of which is loss. We then take the study to a real 30M-parameter DeltaNet-class language model in two steps. First, an inverted control: the pre-registered skill-*creation* gate fails — fine-tuning a pretrained model on tracking-rich data never produces the skill (chance through 15k steps, $\approx 30\times$ the budget at which random initialization succeeds) while loss falls throughout, so *insertion*, not retention, is the LM-scale bottleneck. Second, we therefore *construct* the skill in: adding one inert head (widening verified identity to -0.13% perplexity) installs a toggle automaton that reads out behaviorally at 1.00 for -0.098% perplexity, and the toy fate map reproduces — unexercised continued training suppresses behavioral expression to chance (0.52) while the internal circuit stays exact (1.0) and readout-recoverable (1.0); a 10% exercise trickle prevents it; the mechanism is *interface drift* (the tied-embedding readout moves, not the computation). This connects to the unlearning-robustness finding that suppression need not be removal, with the difference that our ground truth is constructed, so we can *prove* the computation survived rather than infer it — under honest caveats (constructed not learned, linear RNN, trivial skill, white-box recovery). All predictions and kills were pre-registered before any run; post-hoc analyses are labeled; every number regenerates from committed artifacts.

1 Introduction

Deliberately installing a capability in a trained model — and keeping it through subsequent training — is a basic operation the field cannot yet perform reliably. The closest literature edits *facts*: rank-one and mass-editing methods write key–value associations into transformer MLPs [13, 14], and a growing durability literature measures how those edits decay under subsequent fine-tuning [2, 16, 6], finding them fragile. Task-vector arithmetic [10] transplants whole fine-tuning deltas, but post hoc — the merged model is the

end state, not the beginning of more training. What none of these settings has is a *ground-truth circuit*: when performance drops after continued training, there is no way to ask whether the inserted computation was destroyed or merely disconnected from behavior.

We build the setting where that question is answerable. Our companion mechanism study [8] constructed *verified-exact* solutions to hard state-tracking word problems (S_5 among them) in DeltaProduct-style Householder linear RNNs, and showed that from such an initialization, gradient descent grows the additive input path $b_t = W_b e_t$ into a parasite that pulls the model off the exact solution — while deleting that path ($-b$) leaves it stable. Those constructions are, exactly, *implants*: skills inserted by hand, bit-verified at insertion time, whose integrity after any amount of further training can be measured against a known ceiling of 1.0 at $16\times$ the training length. This paper uses them to map the fate of inserted skills under continued training — including training that never exercises the skill at all — under four protection regimes, and then asks the same question at language-model scale, where insertion itself becomes the surprise.

Contributions.

1. **A pre-registered fate map for implanted skills (§3)**: a verified-exact S_5 automaton implanted into a vocab-extended model, continued-trained on mixtures $p \in \{0, 10, 50, 100\}\%$ of its own task vs. a foreign task, under four protection arms, three seeds. The frozen predictions are scored as frozen (one confirmed, three dead); the surviving structure is labeled post hoc.
2. **The protection law (§3.1)**: retention is the *interaction* of removing the additive path with any nonzero exercise — perfect (1.000) at $p \geq 10\%$ under $-b$; lost under the standard architecture at every p including 100%; lost under $-b$ at $p = 0$.
3. **The resurrection (§3.2)**: zeroing W_b at inference restores “dead” exercised implants to exactly 1.000 (6/6 cells, including fully-trainable ones) and frozen-core implants even at zero exercise (3/3); only unprotected-and-unexercised implants truly erode. The resulting taxonomy — protected / concealed / eroded — is a recoverability probe the edit-durability literature lacks.
4. **Warm-start skill-acquisition resistance at LM scale (§4)**: the pre-registered creation gate failed — a TinyStories-pretrained 30M linear-RNN LM fine-tuned on 50% tracking data never acquires a toggle-tracking skill (chance through 15k steps) that the identical harness from random init acquires by step 500, at lower loss throughout: the skill is not on the loss-descent path from the pretrained basin.
5. **The fate map reproduces at LM scale, by construction (§5)**: since fine-tuning cannot insert the skill, we *construct* it into one inert added head of the real LM (widening verified identity) and continue training under a pre-registered grid. Concealment and recovery reproduce — unexercised training suppresses behavior to chance while the internal circuit stays exact and readout-recoverable — via an *interface-drift* mechanism, connecting to unlearning robustness under a constructed-ground-truth caveat.

2 Related work

Edit durability. Cheng et al. [2] and Wen and Zhang [16] measure the decay of ROME/MEMIT-class factual edits [13, 14] under subsequent fine-tuning; Gupta et al. [6] document gradual and catastrophic forgetting at editing scale. All operate on key-value associations in transformer MLPs, with no ground-truth circuit and no recoverability probe; our results suggest their central quantity — performance decay — may conflate erasure with bypass (§3.2). **Task vectors.** Ilharco et al. [10] add fine-tuning deltas post hoc; continued training after grafting is not their setting. **Ground-truth circuits.** Tracr [12] compiles exact algorithms into transformer weights and InterpBench [7] builds semi-synthetic transformers implementing known circuits; both establish exact-construction, and neither subjects the construction to continued training — which is our question. **Unlearning robustness.** That behavioral suppression need not be weight-level removal is established: Deeb and Roger [3] recover “unlearned” facts by fine-tuning on adjacent facts; Xu et al. [17] document reversibility; Lee et al. [11] give a training-free activation-patching depth score

exercise p	none (+ b)	$-b$	freeze core	lr $\times$ 0.1 core
0%	0.20 / 1.00	0.32 / 1.00	0.40 / 0.51	0.45 / 1.00
10%	0.26 / 1.00	1.00 / 1.00	0.40 / 0.51	0.25 / 1.00
50%	0.22 / 1.00	1.00 / 1.00	0.28 / 0.51	0.23 / 1.00
100%	0.25 / 0.36	1.00 / 0.26	0.21 / 0.30	0.24 / 0.19

Table 1: End-of-training **implant integrity** / **foreign-task accuracy** (seed medians; integrity = S_5 accuracy at position 511, chance 0.2; parity chance 0.5). The two facts that survive every cell: only $-b$ with nonzero exercise retains the implant (1.00), and freezing the core costs the foreign task almost entirely (0.506) while still not retaining.

(UDS) that flags knowledge recoverable from internal representations behind a clean-looking output. Our §5 is adjacent, not competing: we do not propose a detector or claim to beat behavioral evaluation. Our difference is that the capability is *constructed* ground truth, so we can *prove* the computation is intact (read the implanted head directly) rather than infer residual knowledge, and we identify a specific mechanism (interface drift) for a specific installation route (weight-space implant under continued pretraining, not a targeted unlearning method). **Plasticity.** Warm-starting is known to cost generalization [1], and deep continual learning loses plasticity over long horizons [4]; our Tier-2 observation is the qualitative extreme — zero acquisition of a capability at $30\times$ budget, from a single pretraining, while loss improves. **Substrate.** The implant instrument, the $-b$ lever, and the parasitic-attractor mechanism come from our companion study [8], built on DeltaProduct-class state tracking [5, 15]; the LM tier uses the 30M DeltaNet-class model of our write-economics study [9].

3 Tier 1: the exact-implant fate map

Setup. The substrate is a DeltaProduct-style Householder linear RNN ($d=112$, $n_h=4$, pure linear per-step readout) with vocabulary extended from 2 to 4 tokens. The implant is the verified-exact S_5 construction of Howe [8] — reflection programs realizing the group generators exactly, written into embedding rows 0/1, the reflection projection, the initial state, and the readout — asserted at initialization to give 1.000 accuracy at position 511 ($16\times$ any training length used here) in every cell before training starts. The foreign task is parity on tokens $\{2, 3\}$ (targets on classes $\{0, 1\}$ of the same head); one task per sequence; training length 32. Continued training draws mixtures $p \in \{0, 10, 50, 100\%\}$ of S_5 sequences (60 epochs, Adam, the companion study’s hyperparameters), under four arms: **none** (standard architecture, additive path present and trainable), $-b$ (additive path removed), **freeze-core** (reflection projection and initial state frozen), **discount** (learning rate $\times 0.1$ on the core). Implant integrity is position-511 accuracy on S_5 ; plasticity is parity accuracy; the additive-path norm $\|W_b e\|$ is logged. Predictions IP1–IP4 and kills KI1–KI3 were frozen before any run.

Frozen verdicts. IP4 — the attractor echo — **confirmed:** at $p=100\%$ under the standard architecture, in-domain accuracy stays 1.0 while integrity collapses to 0.2467 and $\|W_b e\|$ grows from 0 to 0.9901: the companion study’s parasite, operating on an implanted circuit. IP1 (a $\geq +0.30$ $-b$ -over-none margin at $p=0$) died at $+0.127$; IP2 (exercise monotonically protects under the standard architecture) died — the standard arm is uniformly dead (0.1967–0.2567 across all p); IP3 (freeze-core retains ≥ 0.90 everywhere) died (0.4 at best). KI1 held in all 48 cells (implant exactly 1.000 at initialization); KI2 held (parity reaches 1.0 under the default arm); KI3 does not fire. Three of four frozen predictions dead — and Table 1 shows why: the margins were aimed at the wrong cells.

3.1 The protection law is an interaction

The structure the grid actually shows (post hoc, three seeds, Fig. 1): the $-b$ implant retains **perfectly** — integrity 1.0/1.0/1.0 at $p=10/50/100\%$ — while the standard (+ b) implant is destroyed at every exercise rate, *including* full exercise, and the $-b$ implant at zero exercise dies too (0.3233, via shared-readout and projection-leak gradients from the foreign task). Neither lever suffices alone: protection = (no additive

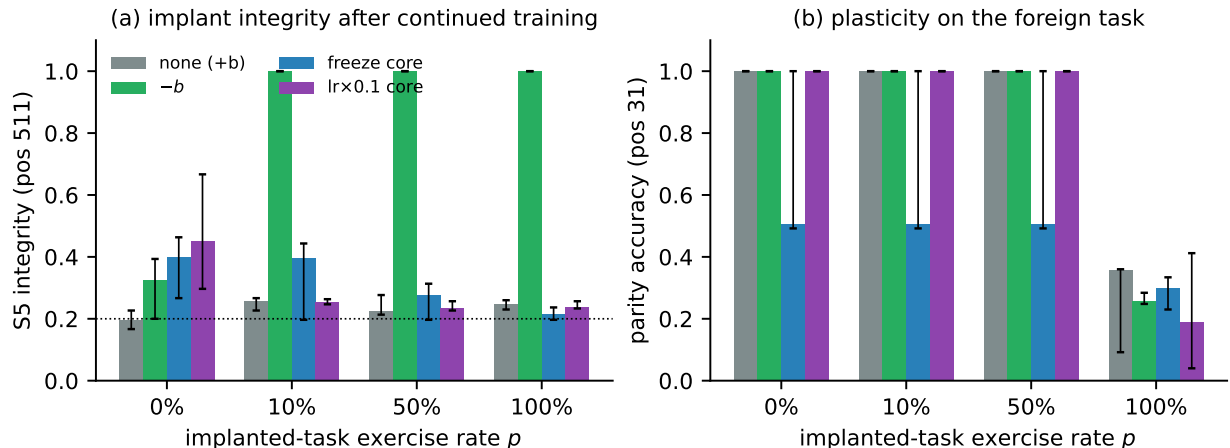


Figure 1: **(a)** End-of-training implant integrity by exercise rate and protection arm (bars: seed medians; whiskers: seed range; dotted line: chance). Only $-b$ with $p \geq 10\%$ retains. **(b)** Foreign-task (parity) accuracy: freezing the core collapses plasticity; every other arm learns the foreign task fully at $p < 100\%$.

bypass) $\times$ (any exercise trickle). Partial levers land in between and pay steeply: freezing the core neither retains (0.4 at best) nor stays plastic (parity 0.506 vs. 1.0 unfrozen — a 0.494 plasticity cost), and a $10\times$ core learning-rate discount is indistinguishable from no protection.

3.2 The resurrection: concealed, not erased

Why does the exercised (+b) implant die even while its task is half the training data? The companion study’s mechanism predicts the answer: it doesn’t. We reran the four decisive conditions ($\{\text{none, freeze-core}\} \times p \in \{0, 100\}\%$, three seeds each; identical configurations; disclosed as post-hoc exploratory) and evaluated end-of-training integrity twice — as trained, and with the additive projection zeroed at inference (Fig. 2):

- $p=100\%$, both arms, all six cells: integrity 0.246/0.216 as trained $\rightarrow$ **exactly** 1.0 with W_b zeroed. The implanted automaton survived 60 epochs of training *intact*; cross-entropy grew a bypass around it ($\|W_{be}\| \approx 0.928\text{--}1.398$) that conceals it behaviorally.
- $p=0\%$, freeze-core, all three cells: $\rightarrow$ 1.0. With the transition circuit frozen, even training that never exercises the skill leaves it fully recoverable.
- $p=0\%$, unprotected, all three cells: NOT recoverable at depth (0.227) although in-domain behavior restores (0.981): with every parameter free and no exercise, the deep automaton structure genuinely erodes.
- The bypass norm scales with exercise ($\|W_{be}\|$: 0.928 at $p=100\%$ vs. 0.162 at $p=0$): loss pressure on the implanted task *feeds* the parasite, exactly the norm-inflation channel of Howe [8].

The fate taxonomy: an implanted skill under continued training ends **protected** ($-b$ + exercise: never harmed), **concealed** (+b + exercise, or frozen core: behaviorally dead, fully recoverable by a one-line intervention), or **eroded** (unprotected *and* unexercised: deep structure lost). Two consequences. For deliberate skill installation, the recoverability question — not the raw-retention question — is the right one, and the additive path is the master switch. For the edit-durability literature, which reports decay without a ground-truth circuit [2, 16], our result is a caution: what is measured as an erased edit may be a concealed one, and the difference matters both for safety (concealed capabilities can be restored) and for method design (protect the bypass, not the weights).

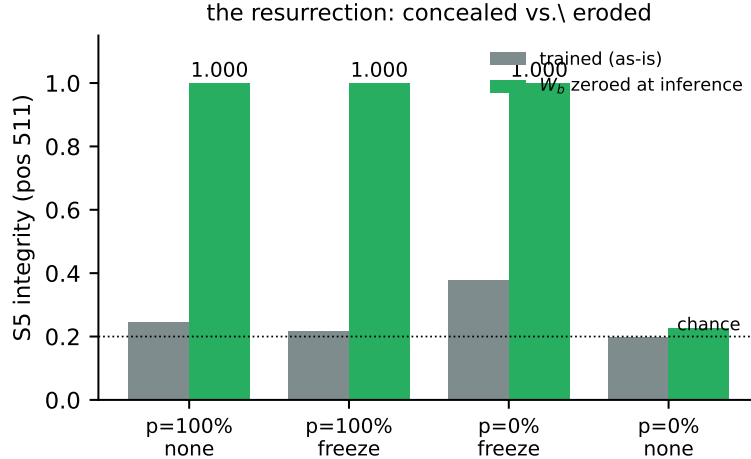


Figure 2: The b-zero reveal on “dead” implants (means over three seeds; post-hoc exploratory): zeroing the additive projection at inference restores exercised implants and frozen-core implants to exactly 1.000; only the unprotected-and-unexercised condition is genuinely eroded.

4 Tier 2: at LM scale, insertion is the bottleneck

The pre-registered LM tier (a 30M-parameter DeltaNet-class TinyStories LM from Howe 9) was designed to measure *retention*: create a toggle-tracking skill by fine-tuning on a 50% tracking mixture, then watch it decay under pure-LM training with four protection arms. It never got there — the frozen creation gate (toggle-8 ≥ 0.90 after 2,500 steps, one disclosed escalation to 5,000) **failed**: the pretrained model sat at exactly chance (0.5, the constant-answer signature) through the full escalation while mixture loss fell normally.

Two post-hoc controls (labeled) sharpen this into a finding (Fig. 3). A **cold control** — the identical harness, data, optimizer, and batch geometry from *random* initialization — acquires the skill by step 500 and holds it (one $n=50$ evaluation flicker aside, visible in Fig. 3). A **warm extension** — the failed checkpoint continued another 10,000 steps (15k total, $\approx 30\times$ the cold emergence time) — stays at chance at all 21 evaluations. And the loss ordering is inverted throughout: the pretrained model has strictly better language quality and lower mixture loss at every step and never acquires the skill; the random init pays perplexity (5.149 vs. 3.082 at the pretrained start) and gets it almost immediately. The skill is not on the loss-descent path from the pretrained basin: the pretrained model can answer the tracking checks with a constant policy and spend its gradient budget polishing everything else, and 300M tokens of pretraining made that basin inescapable at this budget — *warm-start skill-acquisition resistance*, blocked not delayed, qualitative rather than the quantitative slowdowns of the plasticity literature [1, 4].

The program consequence inverts the study’s premise: at LM scale via fine-tuning, the bottleneck for deliberate skill installation is **insertion**, not retention. That elevates Tier 1’s instrument — direct weight-space implantation, where insertion is exact by construction and the protection law governs what happens next — from toy convenience to the load-bearing route, and makes “construct the exact implant inside a real LM layer” the natural next-scale problem.

5 Tier 3: constructing the fate map at LM scale

Since fine-tuning cannot insert the skill (Tier 2), we take the route Tier 1 makes available: *construct* it directly into the trained LM, then continue training and measure its fate. This is pre-registered separately (freeze before the continued-training battery).

The construction obstacle. A trained tied-embedding LM has no spare residual capacity: over TinyStories, the least-active residual dimension has standard deviation 0.717 against a median of 0.87.

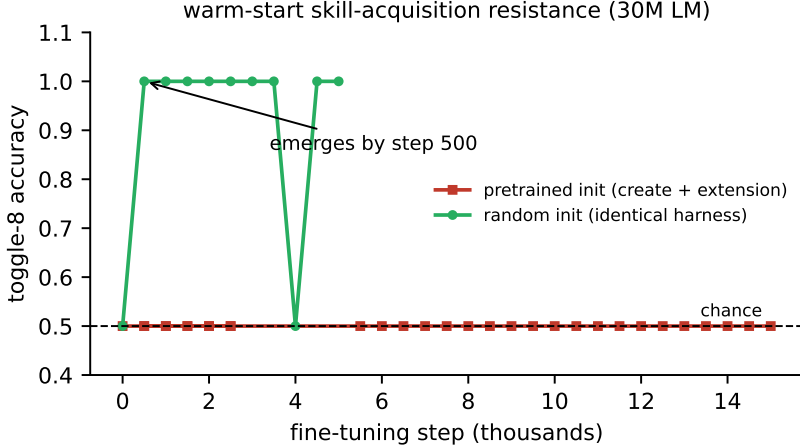


Figure 3: Warm-start skill-acquisition resistance (30M LM, identical fine-tuning harness): a random init acquires toggle tracking by step 500; the TinyStories-pretrained model stays at chance through 15k steps at lower loss throughout.

arm	p	beh. start	beh. end	internal end	resurrection	ppl end
none	0%	1.00	0.52	1.00	1.00	3.41
none	10%	1.00	1.00	1.00	1.00	3.47
none	50%	1.00	1.00	0.45	1.00	3.72
freeze	0%	1.00	0.00	0.00	0.00	3.42
freeze	10%	1.00	1.00	0.45	1.00	3.49
freeze	50%	1.00	1.00	0.76	1.00	3.74

Table 2: Tier-3 continued-training battery (widened LM, seed 0, 3000 steps; toggle chance 0.5). “Internal” reads the implanted head’s own computation; “resurrection” is behavioral toggle after re-pinning the readout interface.

Donating one attention head is cheap but head-specific (Fig. 4a: block-0 head 1 costs 0.3% perplexity, head 6 a catastrophic 431.9%), and routing the circuit’s answer to the logits *in place* collides with the block’s output RMSNorm and drives perplexity to $\approx 10^{14}$. **The widened substrate.** Adding one inert 64-dimensional head (zero-padding every weight) is verified identity (-0.13% perplexity); implanting the toggle circuit into that fresh head is then clean, because the new dimensions are exactly zero for real tokens (Fig. 4b): behavioral toggle 1.00 through depth 64 at -0.098% perplexity. An exact skill lives in a real LM and it reads out, free.

The battery. We continue training the implanted widened model for 3000 steps under a pre-registered grid: arm $\in \{\text{all-trainable}, \text{freeze-implant}\} \times \text{exercise } p \in \{0, 10, 50\}\%$, logging behavioral toggle (logits), internal toggle (the implanted head read directly), and perplexity; after training we re-pin the readout interface and re-measure (the resurrection probe). Table 2 and Figs. 5–6 give the result, and the Tier-1 taxonomy reproduces:

- **Concealment forms under training and recovers.** All-trainable, 0% exercise: behavioral toggle decays to chance (0.52) while the internal circuit ends exact (1.0) and re-pinning the readout restores it (1.0). The computation survives 3000 steps of pure-LM training; only its behavioral *expression* is lost.
- **The mechanism is interface drift.** The dedicated tokens’ embeddings and the readout are tied into the shared embedding and take gradients every step, so continued training drifts the *interface* while the head’s computation persists. A 10% exercise trickle pins the interface (behavior stays 1.00); freezing the circuit’s weights but not its interface is *counterproductive* (all-trainable co-adapts and

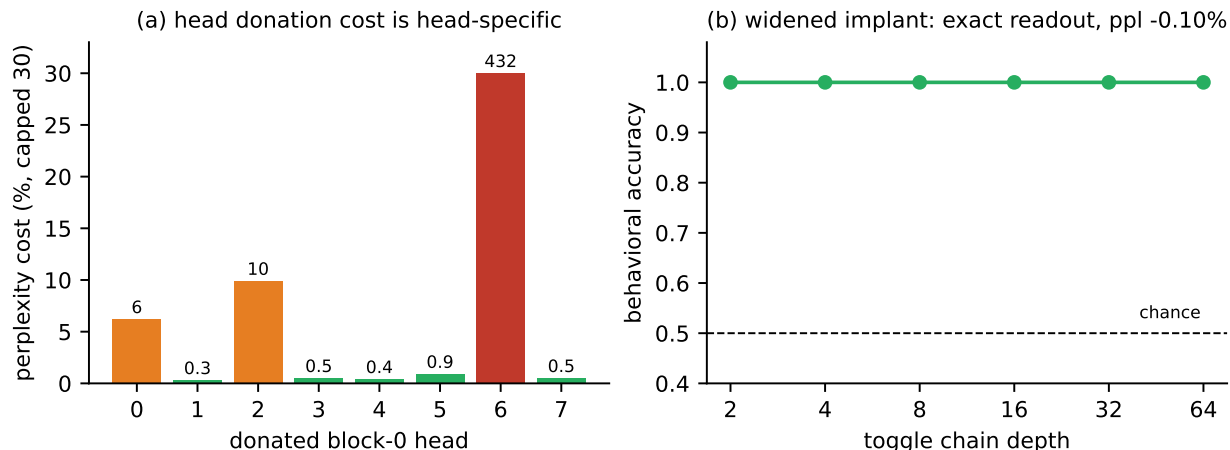


Figure 4: **(a)** Perplexity cost of donating each block-0 head (capped 30%; head 6 is 431.9%). **(b)** In the widened substrate the implanted circuit reads out at 1.00 through depth 64, at -0.098% perplexity.

re-forms, internal 1.0; freeze/0% cannot and genuinely erodes, internal 0.0, unrecoverable 0.0).

- **Skill diffusion.** At 50% exercise behavioral toggle stays 1.00 while the single-head internal read drops (0.45): heavy practice spreads the skill into redundant routes beyond the implanted head.

The concealed/eroded distinction again separates on the internal read (Fig. 6) — which, relative to unlearning-robustness detectors [3, 11], we can here *ground* against a known-intact construction rather than infer.

6 Limitations and disclosures

Tier 1 is toy-scale by design — that is what buys the exact ground truth — with one implant task (S_5), one foreign task (parity on a shared head), one optimizer, and 112 dimensions; the protection law and taxonomy are three-seed toy results whose LM-scale analogues are untested (P4 of Howe 9 suggests the LM regime can differ). Three of four frozen Tier-1 predictions failed as frozen; everything in §3.1–§3.2 beyond IP4 is post-hoc structure requiring pre-registered replication to harden. The reveal reruns duplicate the original cells’ configurations rather than reusing saved weights (checkpoints were not retained; CUDA nondeterminism makes trajectories near- rather than bit-identical). Tier 2 is one model, one skill, one fine-tuning recipe, single-seed, with constant learning rate 3×10^{-4} (matched early-phase scale to the from-scratch grid); “blocked” is scoped to this budget (15k steps) and recipe — curricula, higher learning rates, or partial re-initialization might break the resistance, and distinguishing those is future work, not a counter-claim. **Tier 3** shares Tier 1’s toy-skill scope (toggle, one head, one seed) and adds its own: the widened substrate *adds* capacity rather than implanting into the model as-is (the in-place readout is blocked by the dense-stream RMSNorm, itself reported); the continued-training setup drifts perplexity $+10.7\%$ even at zero exercise (learning rate not tuned for minimal-drift continuation), so absolute perplexity is not a clean cost measurement, though the toggle-integrity findings are within-run relative. **On the unlearning connection:** Tier 3 is a mechanistic case study on a *constructed* skill under generic continued training, not a targeted unlearning method on natural knowledge; that suppression can spare the weights is established [3, 11, 17], and we neither claim that as new nor propose a detector — our added value is the ground-truth handle and the interface-drift mechanism, and the recovery here is white-box, not an adversary acting on a released checkpoint.

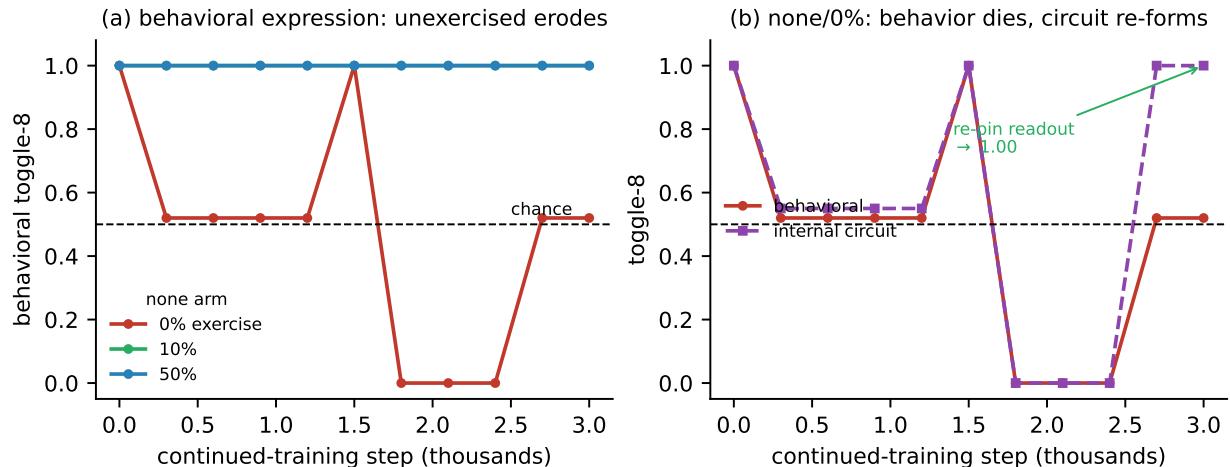


Figure 5: **(a)** Behavioral toggle over continued training (all-trainable arm): unexercised erodes to chance; a 10–50% trickle holds it. **(b)** all-trainable/0%: behavioral expression collapses while the internal circuit — perturbed during training, exact at the end — persists; re-pinning the readout restores behavior.

7 Reproducibility

The full record — pre-registrations (frozen before any run), all Tier-1 cell artifacts, reveal and control artifacts, LM logs, the Tier-3 construction scripts and 6-cell battery, verdicts, and this paper’s regeneration pipeline — is published in the `implant/` and `lminplant/` directories of <https://github.com/gunnerhowe/automaton-underneath> (companion records: `statetrack_note/`, `twogap/`, `proagate/`). Key commits (private working repository; access on request): Tier-1/2 pre-registration freeze 39b2b88, Tier-1 verdicts 7d0d8d8, Tier-2 verdicts 0786a3e; Tier-3 construction 1fd5cf4 and battery pre-registration/verdicts. Every number here is a macro generated by `gen_numbers.py` from committed artifacts (Tier-3 numbers read the `lminplant/ facts.json` and battery cells) and byte-verified by `verify_regen.py`. Total compute: ≈ 5 GPU-hours (Tier 1 + reveal), ≈ 7 (Tier 2), ≈ 2.5 (Tier 3) on one RTX 3080.

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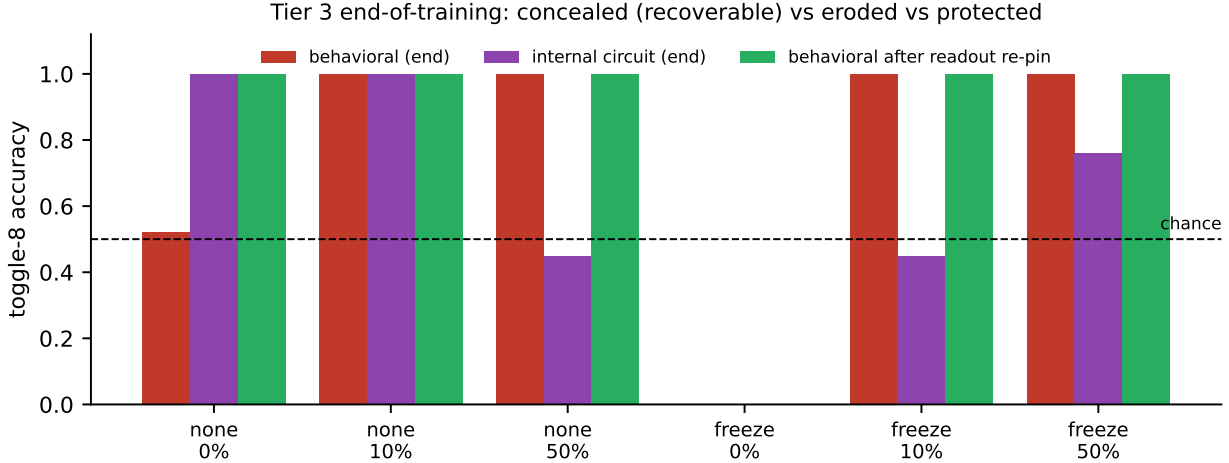


Figure 6: Tier-3 end-of-training state per cell: concealed-but-recoverable (all-trainable/0%), eroded (freeze/0%), and protected (exercised) separate on the internal read, exactly as at toy scale (§3.2).

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